THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · EASY LEVEL UNIT TEST SET 2

Chapters 1–4: **Real Numbers, Polynomials, Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The roots of $(x-2)(x-3)=0$ are: 1

(a) 2 and 3 **(b)** 5 and 6 **(c)** 2 and -3 **(d)** -2 and -3

2. The prime factorisation of 60 is: 1

(a) 2×29 **(b)** $2^2 \times 3 \times 5$ **(c)** 2^6 **(d)** 2×31

3. A polynomial of degree 2 has at most how many zeroes? 1

(a) 3 **(b)** infinitely many **(c)** 1 **(d)** 2

4. The graph of $x = 2$ is a line parallel to the: 1

(a) x-axis **(b)** y-axis **(c)** line $y = x$ **(d)** none of the axes

5. The discriminant of $x^2 - 5x + 6 = 0$ is: 1

(a) 49 **(b)** 13 **(c)** -1 **(d)** 1

6. The decimal expansion of $\frac{3}{8}$ is: 1

(a) terminating **(b)** non-terminating repeating
(c) an integer **(d)** non-terminating non-repeating

7. Which of the following is a zero of $x^2 - 5x + 6$? 1

(a) 2 **(b)** -2 **(c)** 5 **(d)** 1

8. The lines $x + 2y = 3$ and $2x + 4y = 6$ are: 1

(a) coincident **(b)** perpendicular
(c) parallel **(d)** intersecting at one point

9. The nature of the roots of $x^2 - 5x + 6 = 0$ is: 1

(a) two distinct real roots **(b)** three real roots
(c) no real roots **(d)** two equal real roots

10. The HCF of 20 and 30 is: 1

(a) 60 **(b)** 10 **(c)** 20 **(d)** 5

11. If $p(x) = 2x^2 + x - 5$, then $p(3)$ is: 1

(a) 21 **(b)** 16 **(c)** 11 **(d)** 10

12. The pair of equations $x + y = 5$ and $2x + 2y = 7$ has: 1

(a) no solution (parallel lines)
(b) exactly two solutions
(c) exactly one solution (intersecting lines)
(d) infinitely many solutions (coincident lines)

13. Which of the following is a quadratic equation? 1

(a) $x^3 - 5x + 6 = 0$ **(b)** $x - 5 = 6$ **(c)** $x^2 - 5x + 6 = 0$ **(d)** $-5x + 6 = 0$

14. The LCM of 6 and 9 is: 1

(a) 54 **(b)** 15 **(c)** 18 **(d)** 3

15. The zero of the linear polynomial $3x + 9$ is: 1

(a) -3 **(b)** -9 **(c)** 3 **(d)** 9

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the zero of the polynomial $3x - 12$. 2

17. Solve: $x + y = 12$, $x - y = 4$. 2

18. Find the roots of $(x - 4)(x - 5) = 0$. 2

19. Without long division, state whether $\frac{11}{16}$ is terminating or non-terminating. 2

20. Find the zeroes of $x^2 - 7x + 10$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Find the discriminant of $2x^2 - 4x + 3 = 0$ and hence the nature of its roots. 3

22. Express $\frac{3}{16}$ as a decimal after checking its type. 3

23. Verify that 1 is a zero of $x^3 - 6x^2 + 11x - 6$. 3

24. The sum of two numbers is 25 and their difference is 11. Find the numbers. 3

OR

The product of two consecutive positive integers is 20. Find the integers. 3

25. Find the LCM and HCF of 96 and 404 by prime factorisation. 3

OR

Find the zeroes of $x^2 - 4x - 12$ by splitting the middle term and verify the sum of zeroes. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Find the LCM and HCF of 12, 18 and 24 using prime factorisation. 5

OR

Verify the relationship between zeroes and coefficients for the cubic $x^3 - 6x^2 + 11x - 6$ whose zeroes are $1, 2, 3$. 5

27. 2 pencils and 3 erasers cost ₹22, while 3 pencils and 2 erasers cost ₹23. Find the cost of one pencil and one eraser. 5

OR

The length of a rectangular plot exceeds its breadth by 5 m and its area is 66 m². Find its dimensions. 5

END OF QUESTION PAPER **Total: 50 Marks**

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